

King's College London

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BENG/MENG BIOMEDICAL ENGINEERING EXAMINATION

4CCYB061 MATHEMATICS FOR BIOMEDICAL ENGINEERING

Examination Period 1 (January 2018)

TIME ALLOWED: TWO HOURS

PART A: COMPULSORY. ANSWER ALL QUESTIONS. (40 MARKS)

PART B: ANSWER 2 OUT OF 4 QUESTIONS. (30 MARKS)

ANSWER EACH QUESTION ON A NEW PAGE OF YOUR ANSWER BOOK AND WRITE ITS NUMBER IN THE SPACE PROVIDED.

CALCULATORS MAY BE USED. ONLY THE FOLLOWING MODELS ARE PERMITTED:

Casio fx83

Casio fx85.

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PART A: COMPULSORY – ANSWER ALL QUESTIONS (40 MARKS)**Question 1.**

Two vectors a and b are given by $a = i + 2j - k$ and $b = 2i + 3j + k$.

(a) Find the angle between vectors a and b .

[3 marks]

(b) Find a unit vector which is perpendicular to both a and b .

[2 marks]

$$a) \quad \angle \theta = \cos^{-1} \left(\frac{a \cdot b}{|a| \cdot |b|} \right)$$

$$|a| = \sqrt{1^2 + 2^2 + (-1)^2} = \sqrt{6}$$

$$|b| = \sqrt{2^2 + 3^2 + 1^2} = \sqrt{14}$$

$$\underline{a \cdot b} = 2 + 6 + -1 = 7$$

$$\rightarrow \angle \theta = \cos^{-1} \left(\frac{7}{\sqrt{14} \sqrt{6}} \right)$$

dot product

$$= 40.2^\circ$$

$$\text{unit vector} = \frac{a \times b}{|a \times b|} \Rightarrow$$

$$\underline{a \times b} = \begin{vmatrix} i & j & k \\ 1 & 2 & -1 \\ 2 & 3 & 1 \end{vmatrix}$$

$$|a \times b| = \sqrt{(5)^2 + (-3)^2 + (1)^2} = \sqrt{35}$$

$$= i(2 - (-3))$$

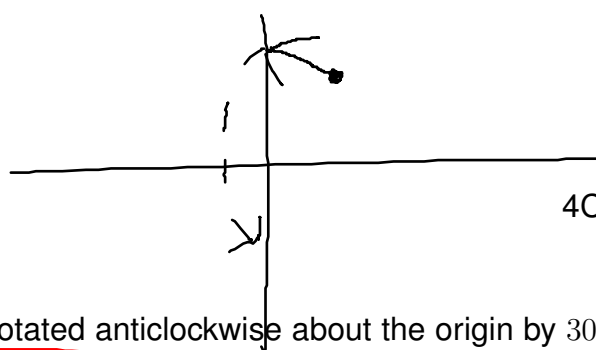
$$- j(1 - (-2))$$

$$- k(3 - 4)$$

$$\rightarrow v = \frac{(5, -3, 1)}{\sqrt{35}}$$

$$= 5i - 3j + k$$

$$= \left(\frac{5}{\sqrt{35}}, \frac{-3}{\sqrt{35}}, \frac{1}{\sqrt{35}} \right)$$



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Question 2.

The point (2, 3) is first rotated anticlockwise about the origin by 30 degrees and then reflected in the x-axis.

- (a) Compute the composite transformation matrix for this rotation followed by reflection.

[3 marks]

- (b) Compute the final location of the point undergoing such a composite transformation.

[2 marks]

a) Rotation matrix $T = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$

$$T = \begin{pmatrix} \cos 30^\circ & -\sin 30^\circ \\ \sin 30^\circ & \cos 30^\circ \end{pmatrix} \Rightarrow \begin{pmatrix} 2 \\ 3 \end{pmatrix} \times \begin{pmatrix} \cos 30^\circ & -\sin 30^\circ \\ \sin 30^\circ & \cos 30^\circ \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} 2 \times \cos 30^\circ + 3 \times -\sin 30^\circ \\ 2 \times \sin 30^\circ + 3 \times \cos 30^\circ \end{pmatrix} = \begin{pmatrix} 0.232 \\ 3.598 \end{pmatrix}$$

Reflection in the x-axis: $T = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

b) $\begin{pmatrix} 0.232 \\ 3.598 \end{pmatrix} \times \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0.232 \\ -3.598 \end{pmatrix}$

Question 3.

Consider the function $f(x, y)$ given by

$$f(x, y) = 3x^2 + xy - 4y$$

- (a) Compute the first partial derivatives of f with respect x and y .

[2 marks]

- (b) Hence, find the stationary points associated with f and the value of the function at these points.

[3 marks]

a)

$$\frac{\partial f}{\partial x} = 6x + y$$

$$\frac{\partial f}{\partial y} = x - 4$$

$$\begin{aligned} 6x + y = 0 &\Rightarrow \begin{cases} 6x + y = 0 \\ 6x - 24 = 0 \\ y + 24 = 0 \\ y = -24 \end{cases} \\ x - 4 = 0 & \end{aligned}$$

When $x = 4$, $y = -24$

$$\begin{aligned} \rightarrow f(4, y) &= 3 \times 4^2 + 4 \times -24 - (-24) \times 4 \\ &\rightarrow x = 4 \end{aligned}$$

$$\begin{aligned} &= 3 \times 16 + -96 + 96 \\ &= 48 \end{aligned}$$



Question 4.

(a) Given that

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

differentiate e^x using the limit definition of the derivative.

a) Recall: $\frac{dy}{dx} = \lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x}$

[5 marks]

$$= \lim_{\delta x \rightarrow 0} \frac{(y + \delta y) - y}{\delta x}$$

$$\rightarrow \frac{d}{dx}(e^x)$$

$$= \lim_{\delta x \rightarrow 0} \frac{e^x + \delta x - e^x}{\delta x}$$

$$= \lim_{\delta x \rightarrow 0} \frac{e^x e^{\delta x} - e^x}{\delta x}$$

$$= \lim_{\delta x \rightarrow 0} \frac{e^x \cdot (e^{\delta x} - 1)}{\delta x}$$

$$= \lim_{\delta x \rightarrow 0} \frac{e^x \left(1 + \delta x + \frac{\delta x^2}{2!} + \frac{\delta x^3}{3!} + \dots + \frac{\delta x^{n-1}}{n!} - 1 \right)}{\delta x}$$

$$= e^x \lim_{\delta x \rightarrow 0} \left(1 + \frac{\delta x}{2!} + \frac{\delta x^2}{3!} + \dots + \frac{\delta x^{n-1}}{n!} \right)$$

$$\text{Since } \delta x \rightarrow 0 \Rightarrow e^x (1 + 0 + 0 + 0 + \dots + 0)$$

$$\rightarrow e^x \cdot 1$$

$$\Rightarrow f'(x) = e^x$$

Question 5.

(a) Find

$$\int \frac{1}{(x^2 - 4x + 3)} dx$$

[3 marks]

(b) Integrate $\ln x$

[2 marks]

$$x^2 - 4x + 3 \quad \Delta = b^2 - 4ac$$

$$x_1 = \frac{-(-4) + \sqrt{4}}{2} = 3 \quad = \frac{(-4)^2 - 4 \cdot 1 \cdot 3}{2} = 16 - 12 = 4$$

$$x_2 = \frac{-(-4) - \sqrt{4}}{2} = 1$$

$$b) \int \ln x$$

$$\int \ln x \cdot 1$$

$$\rightarrow \int \frac{1}{(x-3)(x-1)} dx$$

$$1 = \frac{A}{x-3} + \frac{B}{x-1} \Rightarrow \int \frac{1}{2x-6} - \frac{1}{2x-2} dx$$

$$x=3$$

$$1 = A(x-1) + B(x-3) \quad \ln(2x-6) - \ln(2x-2) + C$$

$$1 = 2A \rightarrow A = \frac{1}{2}$$

$$\ln\left(\frac{2x-6}{2x-2}\right) + C$$

$$x=1$$

$$1 = -2B \rightarrow B = -\frac{1}{2}$$

$$y = mx + c$$

Question 6.

- (a) Calculate the equation of the tangent to $y = 2x^4 - 3$ at the point $(1, -1)$.

[3 marks]

- (b) What is the gradient of the normal to $y = 2x^4 - 3$ at the point $(1, -1)$?

[2 marks]

$$m = \frac{dy}{dx} = 8x^3$$

$$x=1 \rightarrow \frac{dy}{dx} = 8$$

$$y = 8x + c$$

$$-1 = 8 \times 1 + c$$

$$-1 = 8 \times 1 - 9$$

$$\rightarrow y = 8x - 9$$

$$\frac{dy}{dx} = \frac{-1}{8} \rightarrow -1 = \frac{-1}{8} \times 1 + c$$

$$c = -\frac{7}{8}$$

$$f' = 1 + e^x \quad f'' = e^x \quad f''' = e^x$$

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Question 7.

(a) Find the Taylor series for $f(x) = x + e^x$ about $x = 1$.

$$\text{Taylor series} = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(x_0) (x - x_0)^n \quad [5 \text{ marks}]$$

$$= f(x_0) + (x - x_0) f'(x_0) + \frac{(x - x_0)^2}{2!} f''(x_0)$$

$$+ \frac{(x - x_0)^3}{3!} f'''(x_0)$$

$$= (1 + e^1) + (x - 1)(1 + e^1) + \frac{(x - 1)^2}{2!} e^1 + \frac{(x - 1)^3}{3!} e^1 + \dots +$$

Question 8.

- (a) Write the general form of an ordinary differential equation using x as the dependent variable, t as the independent variable, and n as the order.

[2 marks]

- (b) Solve the following equation

$$7x \frac{dy}{dx} - 2y = -\frac{x^2}{y^6}$$

a) $\frac{d^n x}{d^n t} + f\left(x, y, \frac{dx}{dt}, \dots, \frac{d^{n-1} x}{d t^{n-1}}\right) = 0$ [3 marks]

b) $7x \frac{dy}{dx} - 2y = -\frac{x^2}{y^6}$

$$\frac{dy}{dx} - \frac{2y}{7x} = \frac{-x}{7y^6}$$

$$\frac{dy}{dx} - \frac{2}{7x} y = \frac{-1}{7y^6} \cdot x$$

$$y^6 \frac{dy}{dx} - \frac{2}{7x} y^7 = -\frac{1}{7} x$$

PART B: ANSWER 2 OUT OF 4 QUESTIONS. (30 MARKS)**Question 9.**

Answer the following questions.

- (a) A matrix A has an associated eigenvector x . Describe what happens to x when pre-multiplied by A . Hence, define the meaning of the term eigenvalue (given the symbol λ).

[2 marks]

- (b) Use this relationship between A , x and λ to derive the Characteristic Equation

$$|A - \lambda I| = 0$$

where I is the identity matrix.

[2 marks]

- (c) Now consider the specific matrix given by $A = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 2 & 1 \\ -2 & 1 & -1 \end{pmatrix}$

Show that the eigenvalues associated with this matrix are $\lambda_1 = 1$, $\lambda_2 = -1$ and $\lambda_3 = 2$. Hence, find e_1 and e_2 , the eigenvectors associated with eigenvalues $\lambda_1 = 1$ and $\lambda_2 = -1$.

[8 marks]

- (d) Show that

$$A^2 e_1 - A^2 e_2 = e_1 - e_2$$

where A^2 represents $A \times A$. *Hint:* You do not need to perform an explicit calculation.

[3 marks]

$$c) \quad Ax = \lambda x \quad \Leftrightarrow |A - \lambda I| = 0$$

$$\begin{pmatrix} 1 & -1 & 0 \\ 1 & 2 & 1 \\ -2 & 1 & -1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 0$$

$$\begin{pmatrix} 1-\lambda & -1 & 0 \\ 1 & 2-\lambda & 1 \\ -2 & 1 & -1-\lambda \end{pmatrix} = 0$$

10

SEE NEXT PAGE

$$(2-\lambda)(-1-\lambda) \quad 1.1 \quad -$$

Any point R on the plane with position vector r must be $\perp$ with vector n
 $(r-a) \cdot n = 0$



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Question 10.

- (a) A general plane is defined by the vector equation

$$r \cdot n = a \cdot n$$

where r represents the position vector of a general point within the plane, n is the vector normal to the plane, and a represents the position vector of a specific point within the plane.

Derive an expression for the (shortest) distance between the general plane defined above and the origin. You may need to draw a diagram to show your reasoning.

$$r = (1, 1, 2)$$

[3 marks]

- (b) Two planes are defined by

$$|n_1| = \sqrt{1^2 + 1^2 + 2^2} = \sqrt{6}$$

$$|n_2| = \sqrt{1^2 + (-1)^2 + 1^2} = \sqrt{3}$$

$$\begin{aligned} x + y + 2z &= 4 \\ x - y + z &= 2 \end{aligned}$$

$$\begin{aligned} 2z + 3z &= 6 \\ 2y - 2 &= 2 \\ x + y &= 0 \end{aligned}$$

For each of the two planes defined above, compute the distance between the plane and the origin.

$$\begin{aligned} n_1 \cdot n_2 &= 1 \cdot 1 - 1 \cdot 1 + 2 \cdot 1 \\ &= 2 \end{aligned}$$

$$d = \frac{4}{\sqrt{6}} \quad d = \frac{2}{\sqrt{3}}$$

[3 marks]

- (c) Compute the angle between the two planes.

[2 marks]

- (d) Calculate the equation of the line of intersection formed between the two planes. You may write your answer in either vector or Cartesian form.

[5 marks]

- (e) Explain why the line calculated in the part above cannot intersect the plane given by

$$6x + 6y + 12z = 3$$

$$c) \quad \theta = \cos^{-1} \left(\frac{2}{\sqrt{6} \cdot \sqrt{3}} \right) = 61.9^\circ$$

[2 marks]

d)

e) Same normal

vector $\rightarrow$ Parallel

$\rightarrow$ Never intersect

Question 11.

- (a) Show that the sum of the first n positive integers is given by

$$\sum_{k=1}^n k = \frac{n}{2}(n+1)$$

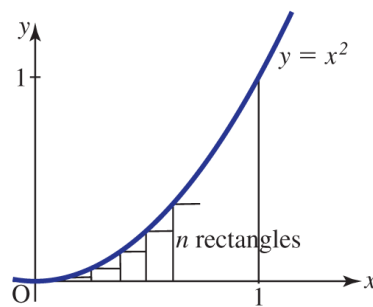
[3 marks]

- (b) Show that the sum of the squares of the first n positive integers is given by

$$\sum_{k=1}^n k^2 = \frac{n}{6}(n+1)(2n+1)$$

[5 marks]

- (c) Find the **exact** area under the graph $y = x^2$ between $x = 0$ and $x = 1$ by dividing the area under the graph into n rectangles, as shown in the figure below.



[6 marks]

- (d) What is the name of the theorem of calculus that relates indefinite integrals (or antiderivatives) to definite integrals?

[1 mark]

Question 12.

- (a) Calculate the area of a circle of radius
- a
- by making use of

$$\int_0^a \sqrt{a^2 - x^2} \, dx,$$

which gives the area of a quarter circle describe by $x^2 + y^2 = a^2$. To solve the integral, make an appropriate substitution. In addition, the trigonometric identity $2 \cos^2 \alpha = 1 + \cos 2\alpha$ may be useful.

[3 marks]

- (b) Calculate the area of a circle of radius
- a
- using the following multiple integral

$$\int_0^a \int_0^{\sqrt{a^2 - x^2}} dy \, dx$$

[2 marks]

- (c) Calculate the area of a circle of radius
- a
- using the following multiple integral in polar coordinates

$$\int_0^{2\pi} \int_0^a r \, dr \, d\theta$$

[2 marks]

- (d) Given a curve with equation
- $y = f(x)$
- , show (using Pythagoras's theorem) that the length of the curve between the points
- $x = a$
- and
- $x = b$
- is given by the formula

$$\text{Length} = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx$$

[3 marks]

- (e) Show that if a curve is defined parametrically by
- $x = x(t)$
- and
- $y = y(t)$
- , then the length of the curve between the points where
- $t = t_1$
- and
- $t = t_2$
- is given by

$$\text{Length} = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \, dt$$

[3 marks]

- (f) Show that the arch length of a complete circle of radius
- a
- is equal to
- $2\pi a$
- .

[2 marks]